

**A PROJECT REPORT
ON
DATA VISUALIZATION WITH PYTHON**



**UNITED
UNIVERSITY**

FACULTY OF COMPUTER APPLICATION

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Grade A

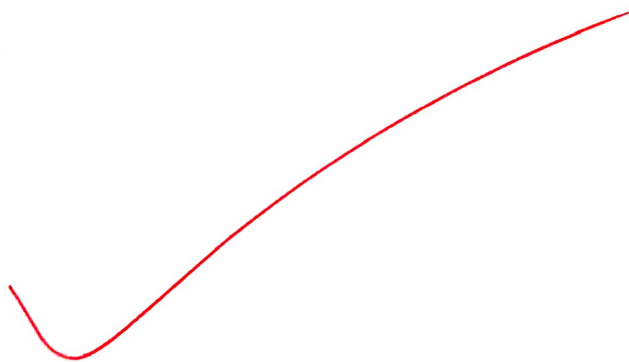
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INTRODUCTION

Data Visualization is an important technique used to represent raw data in a graphical and meaningful format so that patterns, trends, and relationships can be understood easily. In today's digital world, a huge amount of e-commerce data is generated every day through online shopping platforms. Analyzing such large datasets using numerical values alone becomes difficult and time-consuming. Data visualization helps convert complex datasets into clear and understandable visual insights that support business decision-making.

Python is widely used for data analysis and visualization because of its simplicity, flexibility, and powerful libraries. Libraries such as Pandas, NumPy, Matplotlib, and Seaborn allow users to clean, process, analyze, and visualize data efficiently. These libraries help in performing Exploratory Data Analysis (EDA) before applying advanced analytical techniques.

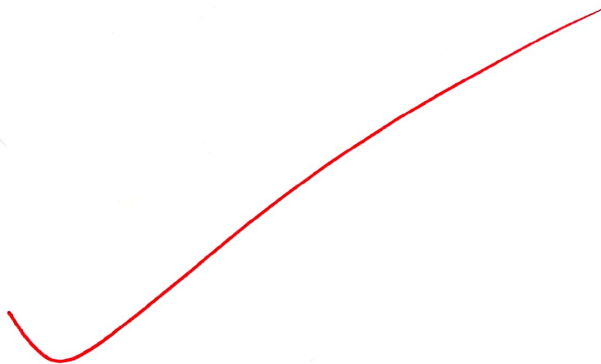
This project focuses on E-Commerce Data Visualization using Python on an online shopping dataset. The dataset contains information related to customer orders, product names, categories, sales amount, profit, payment methods, customer locations, ratings, and monthly sales trends. These attributes help in understanding customer purchasing behavior, best-selling products, and business



OBJECTIVE OF THE PROJECT

The primary objective of this project is to apply data visualization techniques using Python to analyze and understand an E-Commerce sales dataset. The specific objectives of the project are as follows:

1. To understand the structure and characteristics of the E-Commerce dataset using exploratory analysis.
2. To perform Exploratory Data Analysis (EDA) in order to identify patterns, trends, and customer purchasing behaviour from sales data.
3. To analyze customer orders and identify best-selling products based on sales quantity and revenue.
4. To visualize important business factors such as sales amount, profit, payment methods, customer ratings, and monthly sales trends using graphical methods.
5. To study relationships between different variables through bar plots, line graphs, pie charts, histograms, and heatmaps.
6. To analyze different payment methods used by customers and identify the most commonly preferred payment option.
7. To perform customer location analysis in order to understand sales distribution across different cities and regions.
8. To extract meaningful insights from the dataset based on visual observations and graphical analysis.
9. To demonstrate the practical application of Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn in E-Commerce data visualization.
10. To build a foundation for future business analysis, sales prediction, and decision-making using cleaned and well-visualized E-Commerce data.



TOOLS & TECHNOLOGY USED

The following tools and technologies were used to carry out data analysis and visualization in this E-Commerce Data Visualization project:

Python

Python is the primary programming language used in this project due to its simplicity, readability, and powerful support for data analysis and visualization tasks. It provides a wide range of libraries that help in processing, analyzing, and visualizing E-Commerce sales data efficiently.

Jupyter Notebook

The analysis is performed using Jupyter Notebook, which provides an interactive environment for writing Python code, visualizing outputs, and managing datasets efficiently. It allows easy execution of code cells, graphical visualization, and documentation in a single platform.

NumPy

NumPy is used for numerical computations and efficient handling of arrays and mathematical operations required during data preprocessing and analysis.

Pandas

Pandas is used for data loading, cleaning, manipulation, and analysis. It provides powerful data structures such as DataFrames that make E-Commerce data analysis simple and efficient.

Matplotlib

Matplotlib is used for creating basic and customized visualizations such as bar charts, line graphs, histograms, and pie charts to represent sales and customer data visually.

Seaborn

Seaborn is used to create advanced statistical visualizations with attractive designs, including heatmaps, distribution plots, and category-wise comparisons for better understanding of E-Commerce trends.

DATA DESCRIPTION

The dataset used in this project is an E-Commerce Sales Dataset containing information related to customer orders, products, sales, profits, payment methods, customer ratings, and customer locations. The dataset is used to perform data analysis and visualization in order to understand customer purchasing behaviour and business performance.

`df.head()`

	Order ID	Product Name	Category	Sales Amount	Profit	Payment Method	Customer Location	Ratings
0	ORD101	Wireless Earbuds	Electronics	4000	800	UPI	Prayagraj	4.5
1	ORD102	Smart Watch	Electronics	3500	600	Credit Card	Lucknow	4.2
2	ORD103	Running Shoes	Fashion	6000	1200	COD	Kanpur	4.0
3	ORD104	Laptop Backpack	Accessories	2500	500	Net Banking	Varanasi	4.7

In this project, the dataset is analyzed using Python to study sales trends, identify best-selling products, analyze payment methods, and perform profit analysis. The dataset helps in generating meaningful visual insights through different charts and graphs.

The dataset contains both numerical and categorical attributes that are useful for Exploratory Data Analysis (EDA) and visualization techniques. The data is organized in tabular format and each row represents a customer order.

The following are the columns present in the dataset:

1. Order ID – Unique identification number of each order.
2. Product Name – Name of the product purchased by the customer.
3. Category – Category to which the product belongs such as Electronics, Fashion, or Accessories.
4. Sales Amount – Total sales value of the product.
5. Profit – Profit earned from the sale of the product.
6. Payment Method – Payment mode used by the customer such as UPI, Credit Card, COD, or Net Banking.
7. Customer Location – City or region of the customer.
8. Ratings – Customer rating given to the product based on satisfaction.

All these features are either numerical or categorical and are suitable for statistical analysis and visualization.

EDA (EXPLORATORY DATA ANALYSIS).

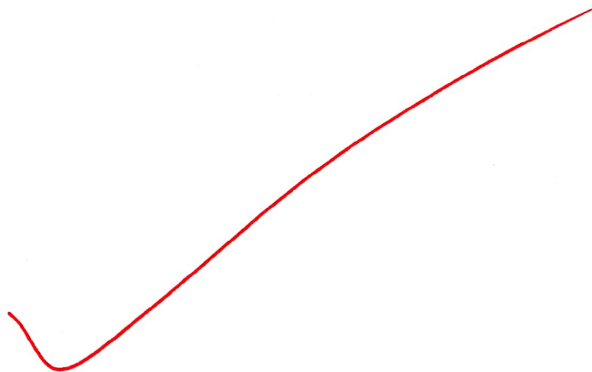
Exploratory Data Analysis (EDA) is an important process used to analyze and understand datasets using statistical and graphical methods. It helps in identifying patterns, trends, relationships, and anomalies present in the data.

In this project, EDA is performed on the

E-Commerce sales dataset to study customer orders, sales performance, profit, payment methods, and customer behaviour. The analysis helps in understanding business trends and customer preferences.

Various visualization techniques such as bar charts, line graphs, pie charts, and heatmaps are used to represent the data visually. These visualizations make the dataset easier to understand and help in identifying meaningful insights.

Overall, EDA provides a strong foundation for data visualization and business analysis using Python libraries such as Pandas, Matplotlib, and Seaborn.



IMPORTING REQUIRED LIBRARIES

```
# Importing Required Libraries
```

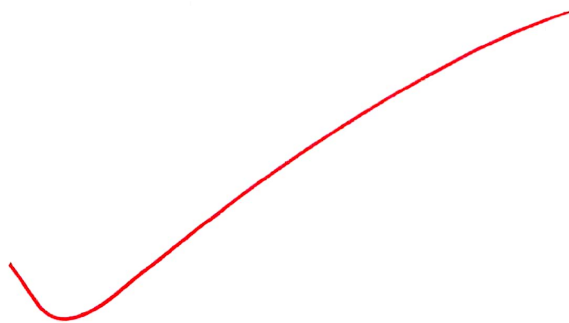
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

The above code imports the required Python libraries used for data manipulation, numerical computation, and data visualization. Pandas and NumPy are used for handling and processing the dataset, while Matplotlib and Seaborn are used to create graphical visualizations.

READING THE DATASET

	Order ID	Product Name	Category	Sales Amount	Profit	Payment Method	Customer Location	Ratings
0	ORD101	Wireless Earbuds	Electronics	4000	800	UPI	Prayagraj	4.5
1	ORD102	Smart Watch	Electronics	3500	600	Credit Card	Lucknow	4.2
2	ORD103	Running Shoes	Fashion	6000	1200	COD	Kanpur	4.0
3	ORD104	Laptop Backpack	Accessories	2500	500	Net Banking	Varanasi	4.7

The dataset is loaded into a Pandas DataFrame using the `read_csv()` function. Displaying the first few rows helps in understanding the structure of the data and the available features.



DATASET INFORMATION

```
# Data overview  
print("Data Info: ")  
df.info()
```

Data Info:

<class 'pandas.core.frame.DataFrame'>

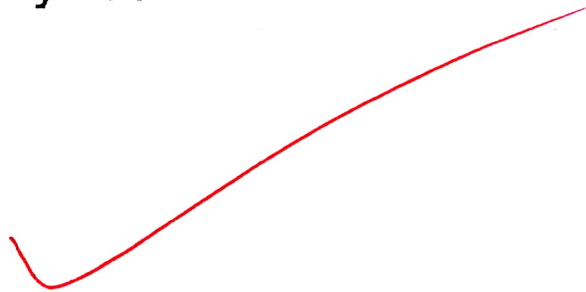
RangeIndex: 4 entries, 0 to 3

Data columns (total 8 columns):

#	Column	Non-Null Count	Dtype
0	Order ID	4 non-null	object
1	Product Name	4 non-null	object
2	Category	4 non-null	object
3	Sales Amount	4 non-null	int64
4	Profit	4 non-null	int64
5	Payment Method	4 non-null	object
6	Customer Location	4 non-null	object
7	Ratings	4 non-null	float64

dtypes: float64(1), int64(2), object(5)

memory usage: 388.0+ bytes



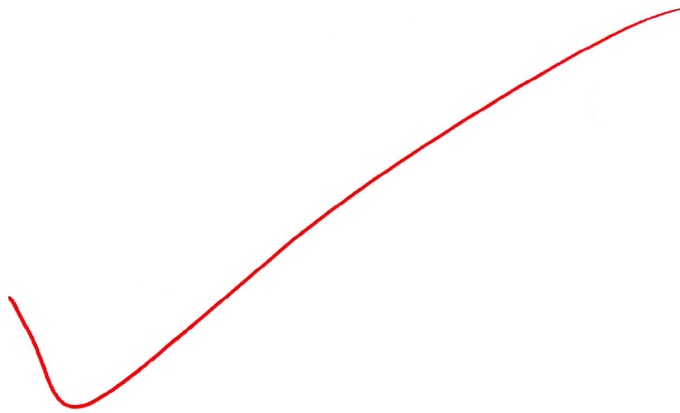

```
print("Missing values:")  
df.isnull().sum()
```

Missing values:

	0
Order ID	0
Product Name	0
Category	0
Sales Amount	0
Profit	0
Payment Method	0
Customer Location	0
Ratings	0

dtype: int64

The dataset consists of E-Commerce sales records containing information related to customer orders, product categories, sales amount, profit, payment methods, customer locations, and ratings. All important attributes are properly populated without missing values, indicating good data quality. The dataset is reliable and well-suited for Exploratory Data Analysis (EDA) and data visualization.

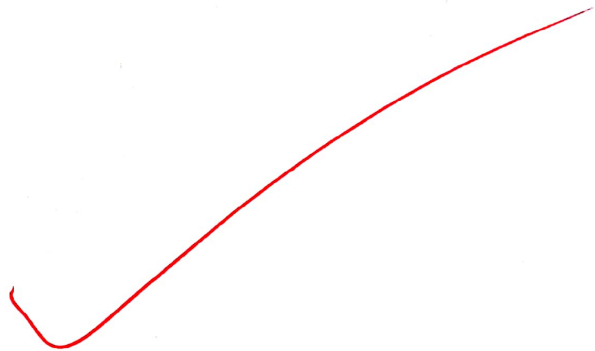


STATICAL SUMMARY OF THE DATASET

STATICAL SUMMARY OF THE DATASET

	Sales Amount	Profit	Ratings
count	4.000000	4.000000	4.000000
mean	4000.000000	775.000000	4.350000
std	1471.960144	309.569594	0.310913
min	2500.000000	500.000000	4.000000
25%	3250.000000	575.000000	4.150000
50%	3750.000000	700.000000	4.350000
75%	4500.000000	900.000000	4.550000
max	6000.000000	1200.000000	4.700000

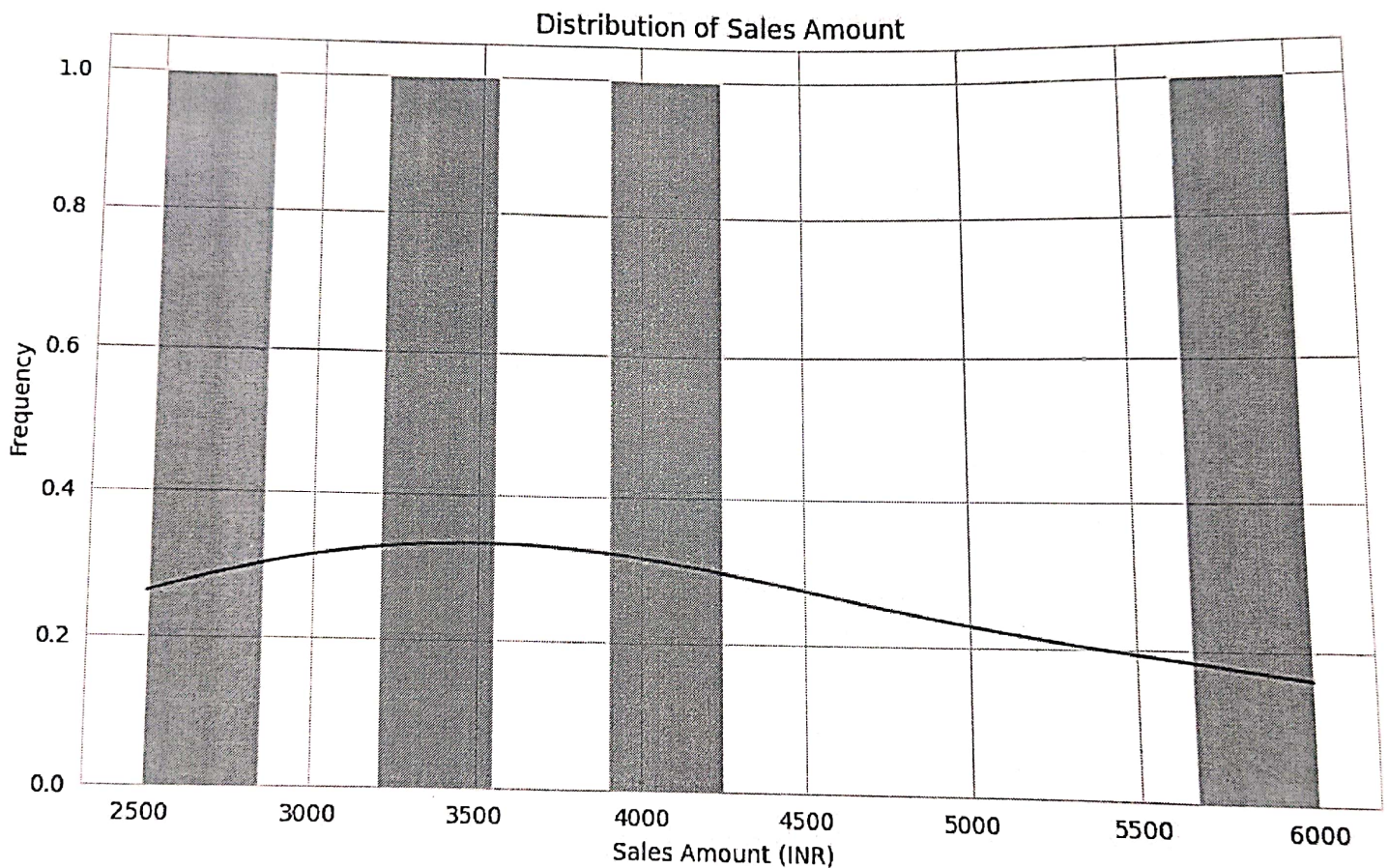
The statistical summary shows important numerical values such as mean, minimum, maximum, standard deviation, and quartiles for the E-Commerce dataset. It helps in understanding sales, profit, ratings, and overall data distribution for better analysis and visualization.



INSIGHTS AND VISUALIZATION

INSIGHTS 1: SALE AMOUNT DISTRIBUTION

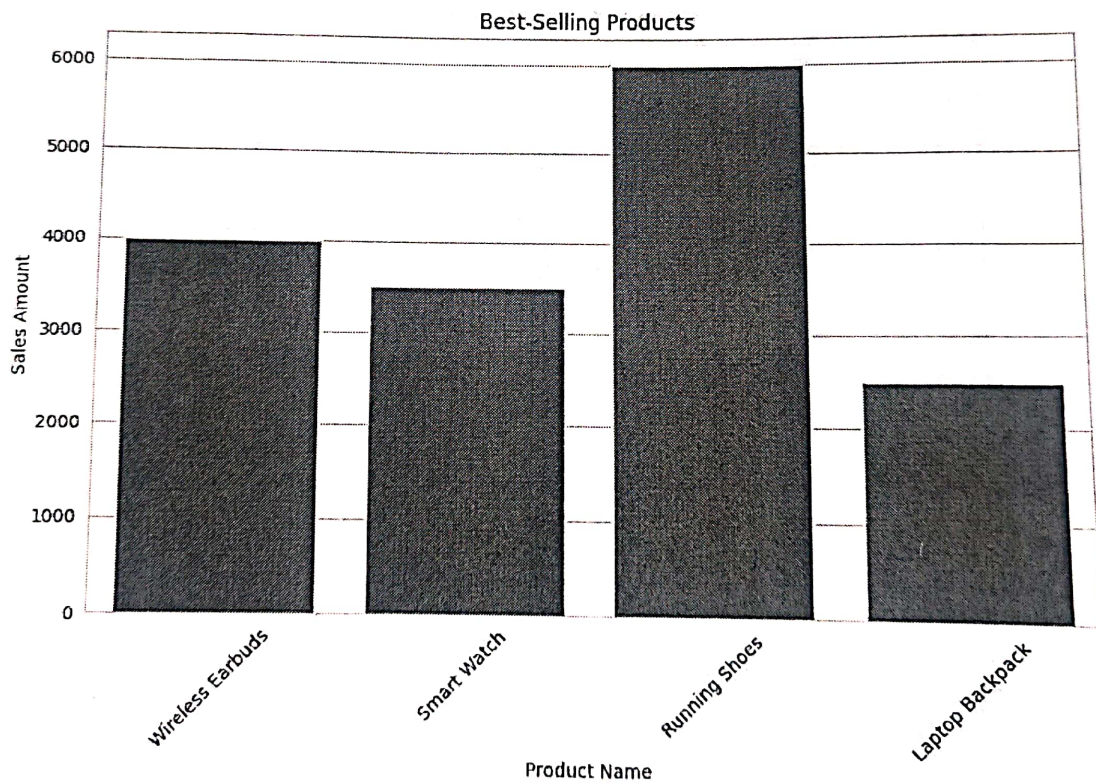
```
plt.figure(figsize=(10,6))  
sns.histplot(df['Sales Amount'], bins=10, kde=True)  
plt.title('Distribution of Sales Amount')  
plt.xlabel('Sales Amount')  
plt.ylabel('Frequency')  
plt.show()
```



The distribution analysis shows that sales amounts range from 2,500 to 6,000 INR. The data follows a uniform frequency where each captured price point occurs once, with the highest density of transactions concentrated between 3,000 and 4,000 INR. While there is a significant activity gap between 4,500 and 5,500 INR, the presence of a sale at the 6,000 INR mark indicates potential for high-value transactions despite the majority of volume sitting at lower price points.

INSIGHTS 2: BEST SELLING -PRODUCTS

```
plt.figure(figsize=(10,6))  
  
sns.barplot(  
    x='Product Name',  
    y='Sales Amount',  
    data=df  
)  
  
plt.title('Best-Selling Products')  
  
plt.xticks(rotation=45)  
  
plt.show()
```

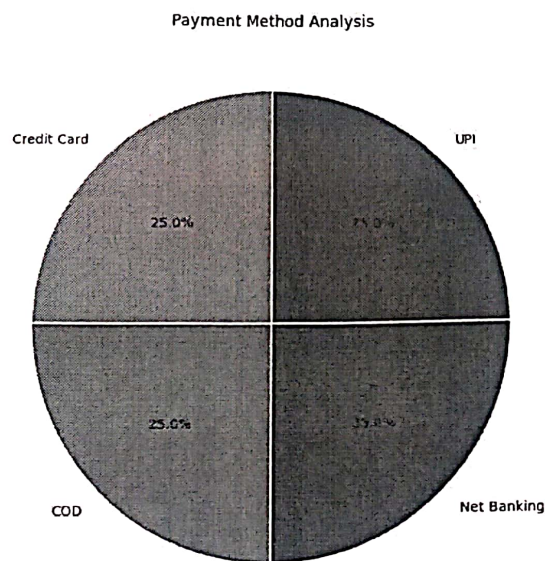


The bar chart identifies Running Shoes as the top-performing product, generating the highest sales revenue at approximately 6,000 INR. It is followed by Wireless Earbuds (~4,000 INR) and Smart Watches (~3,500 INR), while the Laptop Backpack accounts for the lowest sales volume at roughly 2,500 INR. Overall, the data highlights a clear consumer preference for footwear and wearable tech, which significantly outpace accessory sales in this dataset.

INSIGHTS 3: PAYMENT METHOD ANALYSIS

```
df['Payment Method'].value_counts().plot(
    kind='pie',
    autopct='%1.1f%%',
    figsize=(8,8)
)

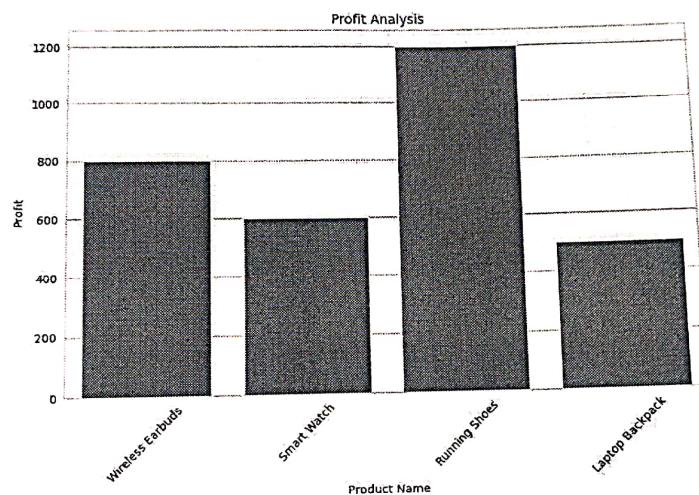
plt.title('Payment Method Analysis')
plt.ylabel("")
plt.show()
```



The pie chart illustrates a perfectly uniform distribution across the four utilized payment channels: UPI, Credit Card, COD (Cash on Delivery), and Net Banking. Each method accounts for exactly 25.0% of the total transactions, indicating that customers do not show a specific preference for any single payment mode. For the project, this suggests a highly balanced payment ecosystem where digital methods (UPI, Credit Card, Net Banking) and traditional methods (COD) are equally integrated and accepted by the consumer base.

INSIGHT 4: PROFIT ANALYSIS

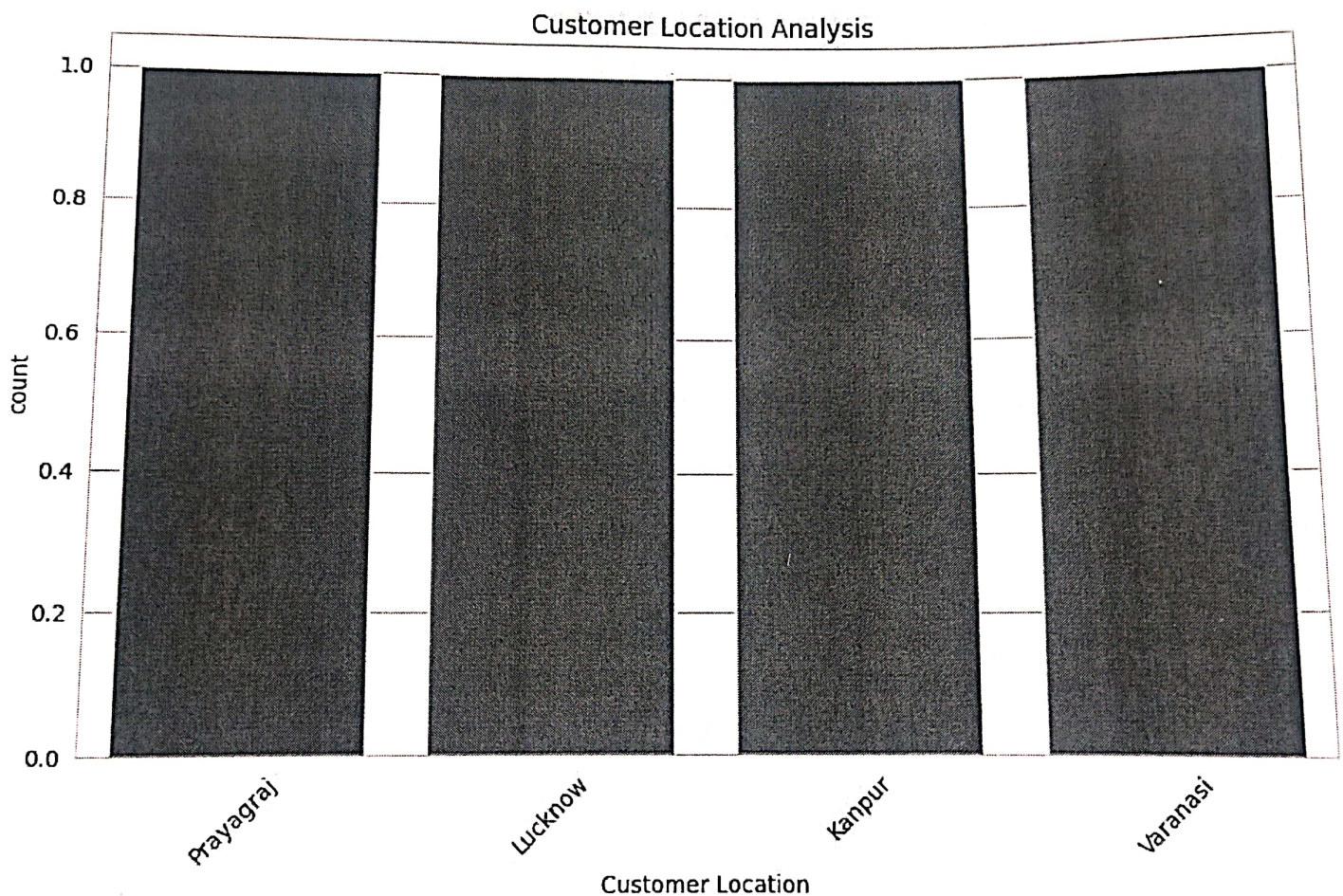
```
plt.figure(figsize=(10,6))  
sns.barplot(  
    x='Product Name',  
    y='Profit',  
    data=df  
)  
plt.title('Profit Analysis')  
plt.xticks(rotation=45)  
plt.show()
```



The bar chart identifies Running Shoes as the most profitable product, yielding a profit of 1,200. It is followed by Wireless Earbuds at 800 and Smart Watches at 600, while the Laptop Backpack contributes the lowest profit at 500. This analysis highlights that Running Shoes are the primary profit driver for the business, significantly outperforming the other categories in the dataset.

INSIGHT 5: CUSTOMER LOCATION ANALYSIS

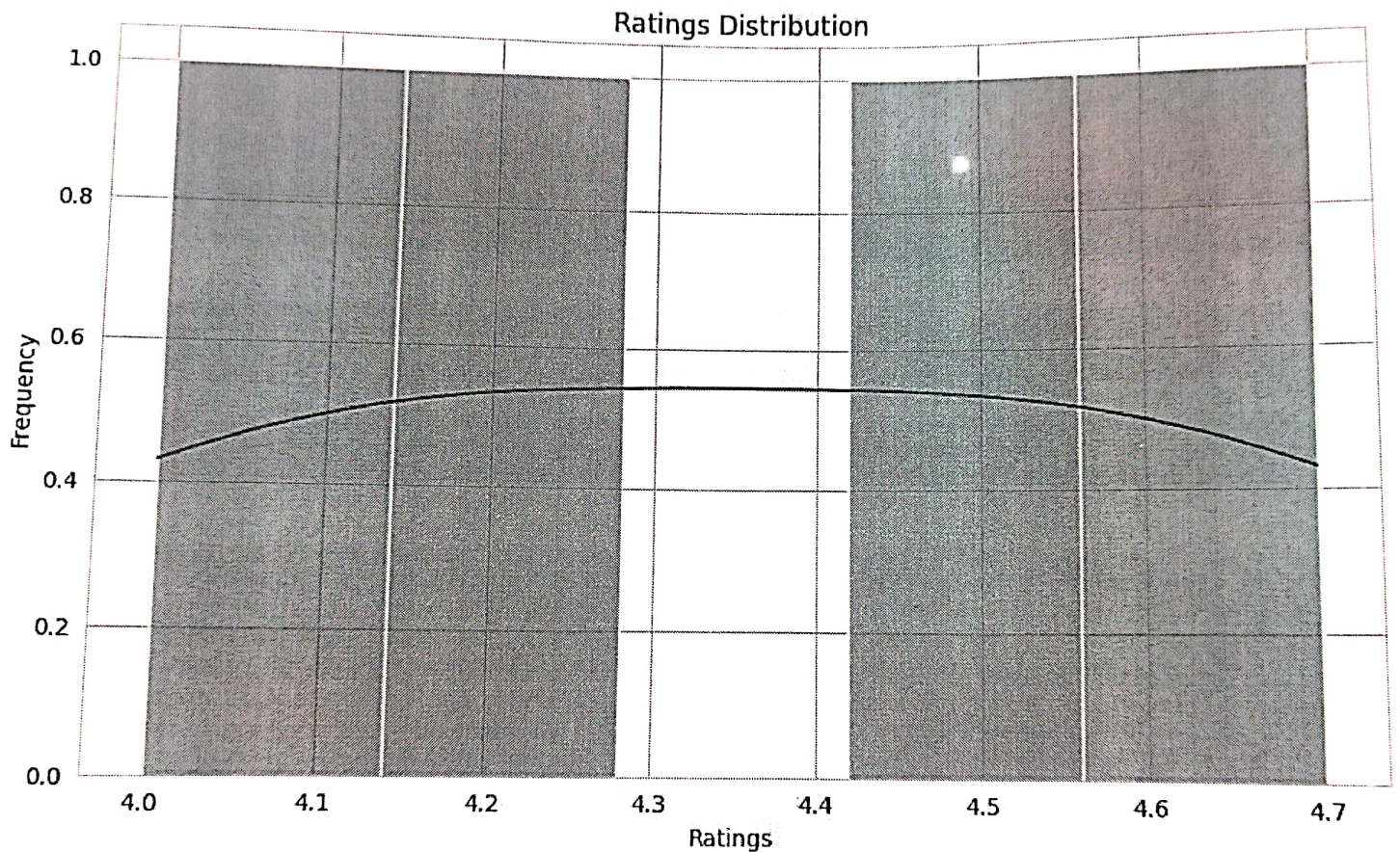
```
plt.figure(figsize=(10,6))  
sns.countplot(  
    x='Customer Location',  
    data=df  
)  
  
plt.title('Customer Location Analysis')  
plt.xticks(rotation=45)  
plt.show()
```



The count plot reveals a perfectly balanced distribution of customers across four cities: Prayagraj, Lucknow, Kanpur, and Varanasi. Each location shows an equal count of 1.0, indicating that the current dataset represents an even geographic spread without a dominant market. For your project, this suggests the business has an equal customer footprint across these key regional hubs, providing a stable foundation for localized marketing strategies in each city.

INSIGHT 6: RATINGS ANALYSIS

```
plt.figure(figsize=(10,6))  
sns.histplot(df['Ratings'], bins=5, kde=True)  
plt.title('Ratings Distribution')  
plt.xlabel('Ratings')  
plt.ylabel('Frequency')  
plt.show()
```



The histogram analysis in highlights customer satisfaction levels, showing that all product ratings fall within a high-quality range of 4.0 to 4.7. The data is distributed evenly with a frequency of 1.0 across most observed rating bins, though a slight gap exists around the 4.3 to 4.4 mark. The Kernel Density Estimate (KDE) curve peaks centrally, suggesting that while feedback is diverse, the overall sentiment is consistently positive and concentrated in the mid-4 range.

INSIGHT 7: SALES VS PROFIT ANALYSIS

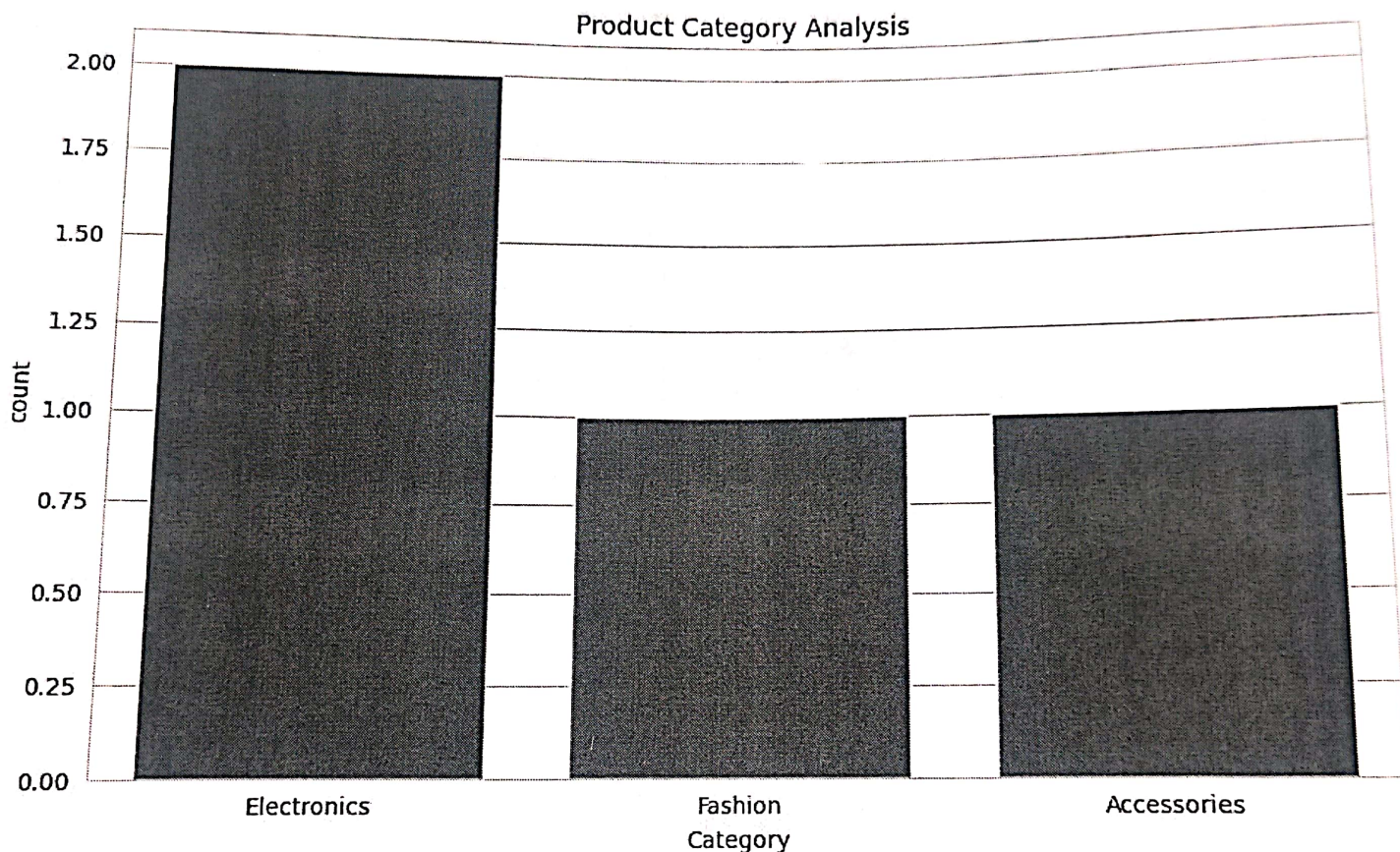
```
plt.figure(figsize=(10,6))  
sns.scatterplot(  
    x='Sales Amount',  
    y='Profit',  
    data=df  
)  
plt.title('Sales vs Profit')  
plt.show()
```



The scatter plot reveals a strong, direct positive correlation between Sales Amount and Profit. As the sales amount increases from 2,500 to 6,000, the profit rises proportionally from 500 to 1,200. This linear relationship indicates a consistent profit margin across the product range, suggesting that higher-priced items are successfully contributing more to the bottom line without a significant increase in relative costs.

INSIGHT 8: CATEGORY ANALYSIS

```
plt.figure(figsize=(10,6))  
sns.countplot(  
x='Category',  
data=df  
plt.title('Product Category Analysis')  
plt.show()
```



The count plot identifies Electronics as the leading product category, with a count of 2.0, doubling the volume of the other segments. The Fashion and Accessories categories show equal representation with a count of 1.0 each. This distribution indicates that the dataset is primarily driven by electronic products, which appear to be the most common offerings or high-demand items compared to the fashion and accessory lines.

INSIGHT 9: CORRELATION HEATMAP

```
plt.figure(figsize=(8,6))

sns.heatmap(

    df.corr(numeric_only=True),

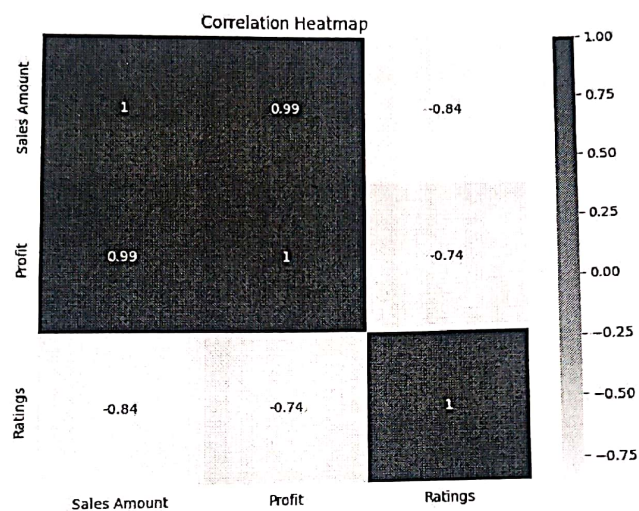
    annot=True,

    cmap='Blues'

)

plt.title('Correlation Heatmap')

plt.show()
```



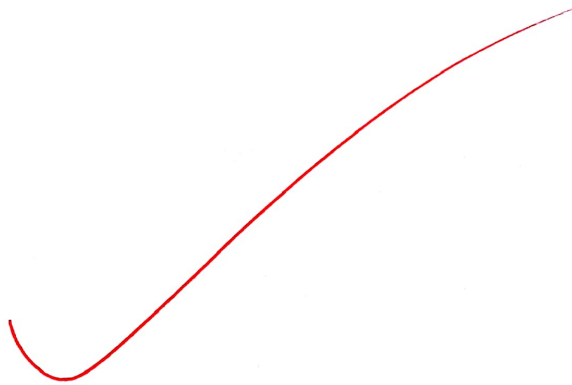
The correlation heatmap illustrates the statistical relationships between the dataset's numerical variables. A nearly perfect positive correlation of 0.99 exists between Sales Amount and Profit, confirming that higher sales lead directly to higher profitability. Interestingly, the analysis reveals a strong negative correlation between Ratings and both Sales Amount (-0.84) and Profit (-0.74). This suggests that in this specific data sample, higher-priced and more profitable items actually received lower customer ratings compared to more affordable products.

CONCLUSION

This project demonstrates the use of Python for E-Commerce data visualization and analysis. Different visualization techniques were applied to study sales amount, profit, payment methods, customer locations, and best-selling products. The graphical representation of data helped in understanding customer purchasing behaviour and business trends effectively.

Exploratory Data Analysis (EDA) was performed to identify important patterns, trends, and relationships present in the dataset. Visualizations such as bar charts, pie charts, line graphs, histograms, and heatmaps made the dataset easier to understand and interpret.

Python libraries such as Pandas, Matplotlib, and Seaborn were successfully used for data analysis and visualization. Overall, the project provided practical knowledge of E-Commerce data analysis and demonstrated the importance of data visualization in business decision-making.



REFERENCE

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